

Public SRA Data Download and Running RNA-Seq SOP

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1 Introduction

This document outlines the procedure for downloading and analyzing RNA sequencing data from public repositories through the Virtual Advanced Computing Cluster (VACC).

This standard operating procedure applies to all users downloading data from public databases on the VACC cluster, whether using macOS, Windows, or Linux.

This SOP provides a step-by-step guide for accessing the cluster, downloading the required data, and organizing it using command-line tools. The data will then be converted into the appropriate formats, followed by quality assessment to ensure its integrity. Once validated, the data will be prepared for downstream bioinformatic analysis.

By following this SOP, users will successfully retrieve and preprocess datasets, ensuring they meet the necessary standards for further analysis. There will be use of basic in-line commands to interact with the operating system. If unfamiliar with these bash commands, there will be a short glossary at the end of this document with helpful references.

This SOP will follow the sequential steps: Connecting to the VACC, loading a work environment, creating a data loading script for raw data acquisition, following the Nextflow pipeline, and finally using Salmon for quantification.

1.1 Picking a Dataset

Choosing an appropriate dataset of high relevance and quality is crucial towards the impact of downstream analysis. The NCBI SRA is the largest repository of raw RNA data. When using this site, a genome sequence of interest can be found and identified through keywords such as disease, species, assay type etc... Once a data set is found, identifying the individual SRR accession codes and storing them in a list will be of later use. The dataset used to create this SOP was the NCBI SRA bioproject PRJNA389279.

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2 Connecting to the VACC Cluster

Note: In order to connect to the VACC cluster, users must be connected to the UVM Wi-Fi or a UVM VPN. Ensure you are on the UVM network before following the steps below or VACC access will be denied.

To download and process the RNA-seq data, users must first establish a connection to the VACC cluster which houses the computing resources required for large-scale data analysis. The following steps outline how to connect to the cluster from varying operating systems.

2.1 Mac, Linux, and Windows

Users on macOS, Linux, and Windows (via Command Prompt, PowerShell, or Windows Terminal) can connect using the following steps: ssh

1. Open the terminal:

- **Mac:** Open **Terminal** (Applications → Utilities → Terminal).
- **Linux:** Open **Terminal** (Ctrl + Alt + T).
- **Windows:** Open **Command Prompt**, **PowerShell**, or **Windows Terminal** from the Start Menu.

2. Type the following command and press **Enter**:

```
ssh YourNetID@login.vacc.uvm.edu
```

3. **Accept the Connection:** The first time you connect, you may receive a warning about the authenticity of the host. You will see a message like:

```
The authenticity of host 'login.uvm.edu (XXX.XXX.XXX.XXX)'  
can't be established.  
ECDSA key fingerprint is SHA256:XXX.  
Are you sure you want to continue connecting (yes/no)?
```

Type **yes** and press **Enter** to proceed. This will add the cluster's server to your list of known hosts. This step only occurs the first time you connect.

4. Enter your UVM netID password when prompted.

Congratulations! You have now successfully entered the Cluster and are ready to begin downloading and preprocessing data.

3 Work Environment

3.1 Module Loading

To work with RNA-seq data, you'll need to load the appropriate software modules. Follow these steps to load the necessary modules for downloading and analyzing the data.

1. First, search for the SRA module by typing the following command into the terminal:

```
module spider sra
```

This command will search for all available modules related to SRA (Sequence Read Archive).

2. Once the search results appear, the system will likely first request the load of `gcc/13.3.0-xp3epyt`, a compiler module required for the SRA tools. You can proceed with the following command to load it:

```
module load gcc/13.3.0-xp3epyt
```

3. After loading the GCC module, load the SRA Toolkit module by running:

```
module load sratoolkit/3.0.0-y2rspiu
```

4. To verify that the modules have been successfully loaded, use the following command to list the currently loaded modules:

```
module list
```

This will display a list of all active modules, confirming that both `gcc/13.3.0-xp3epyt` and `sratoolkit/3.0.0-y2rspiu` have been loaded.

3.2 Preparing Environment

1. Before proceeding, check what directories are available to you by typing:

```
ls #list files
```

Look for a directory named **scratch**, as this is where temporary files should be stored during analysis.

2. Navigate to scratch by using the following command:

```
cd scratch
```

3. After switching directories, verify your new location using:

```
pwd
```

This should return a path ending in **/scratch**, confirming you are now working in the correct location.

4 Creating a Data Loading File

Now that the environment has been prepared, it is time to create a SLURM batch script to automate the download and conversion of RNA seq data from the NCBI's Sequence Read Archive (SRA).

4.1 Writing the Script

The script needs to download the data through its accession name, create a folder under the working directory, and turn it into a fast-q file user fasterq-dump.

1. Begin by creating a working directory where your downloaded datasets will be stored:

```
mkdir directory_name #creates a file called "directory_name"
cd directory_name #enters into the directory_name file
```

2. Open a text editor (e.g., **nano** or **vim**) to create your SLURM batch script:

```
nano script_name.sh #creates script called script_name
```

3. Paste the following script into the editor. This template includes SLURM directives for job scheduling and a customizable list of accession identifiers. The **#SBATCH** identifiers will need to be adjusted for each specific script job according to what is needed from the cluster.

```
#!/bin/bash

#SBATCH --partition=general
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --mem=32G
#SBATCH --time=360:00
#SBATCH --job-name=rnaseq_download
#SBATCH --output=download_output.log
#SBATCH --mail-user=your.email@uvm.edu
#SBATCH --mail-type=ALL

# Define an array of accession numbers (e.g., SRR, ERR...)
list=('ACCESSION1' 'ACCESSION2' 'ACCESSION3')

# Loop through each accession, create a directory, and
# process.
for accession in "${list[@]"; do
    mkdir -p $accession
    cd $accession
    prefetch $accession
    fasterq-dump $accession
```

```
cd ..  
done
```

Replace `ACCESSION1`, `ACCESSION2`, etc., with the appropriate accession numbers for your study.

4. Save and close the file. In `nano`, press `Ctrl + X`, then `Y`, then `Enter`.

4.2 Submitting the Job to SLURM

Once your script is ready, submit it to the cluster using the SLURM job scheduler:

```
sbatch script_name.sh
```

The script will run each download job sequentially, saving the output of each run to its corresponding folder. Progress and status messages will be saved to `download_output.log`, which can be monitored to confirm successful retrieval and to troubleshoot any failed downloads.

4.3 Job Completion

Once the SLURM job is completed, an email will be sent to the email designated within the script. Inside the created working directory "`directory_name`", there should now be the script file "`script_name.sh`", the output log "`download_output.out`", and folders named after each accession number specified within the script.

Note: If you open your created folders and they are empty, or the script failed, you will need to delete the folders created from the SLURM before trying again or the script will fail because it can not create another directory with the same file name.

5 Nextflow Pipeline and Salmon Quantification

Now, the fastq files have been downloaded and stores as one folder per accession into one project folder. Now, the Nextflow pipeline will be used to automate and parallel the RNA-seq pipeline. This will be the connection step from the raw data to quantification using Salmon.

Before beginning, make sure that all files created thus far are under the same data folder. Then, rename all the fastq folders so they are clearly labeled by sample name. Example:

```
mv data/SRR123456_1.fastq.gz data/sample1_1.fastq.gz  
mv data/SRR123456_2.fastq.gz data/sample1_2.fastq.gz
```

This must be done for all samples (must be consistent: `_1.fastq.gz` and `_2.fastq.gz` for paired-end reads).

5.1 Preparing the Salmon Environment

When working on the VACC, modules like nextflow and salmon should already be installed, and just need to be loaded before use. This can be done by typing the following command:

```
module load salmon
```

We must now create a Salmon transcriptome index as a reference. These can be found on Gencode or Ensembl. Example:

```
wget https://ftp.ensembl.org/pub/release-110/fasta/homo_sapiens/cdna/  
Homo_sapiens.GRCh38.cdna.all.fa.gz  
gunzip Homo_sapiens.GRCh38.cdna.all.fa.gz
```

Now, the Salmon index can be created into a folder called "salmon_index":

```
salmon index -t Homo_sapiens.GRCh38.cdna.all.fa -i salmon_index
```

5.2 Preparing the Nextflow Environment

Begin by creating a file to house the Nextflow script. Example:

```
nano rnaseq_salmon.nf
```

Next, copy and paste this script into that nano file. This script tells Nextflow how to process the files you are inputting.

```
#!/usr/bin/env nextflow  
  
params.reads = "data/*_1.fastq.gz"  
params.index = "salmon_index"  
params.outdir = "results"  
  
Channel  
    .fromFilePairs(params.reads, flat: true)  
    .set { read_pairs }  
  
process SalmonQuant {  
    tag "$sample_id"  
    input:  
    tuple val(sample_id), path(reads)  
    output:  
    path("${sample_id}_quant")  
  
    script:  
    """  
    salmon quant -i $params.index -l A \  
    -1 ${reads[0]} -2 ${reads[1]} \  
    """  
}
```

```

        -p 8 -o ${sample_id}_quant
    """
}

```

Next, create a Nextflow configuration file. Example:

```
nano nextflow.config
```

In this file, tell Nextflow how to use SLURM on the VACC. Copy and paste this into the configuration nano script.

```

process {
  executor = 'slurm'
  memory = '32 GB'
  cpus = 8
  time = '24h'
}

params {
  reads = "data/*_1.fastq.gz"
  index = "salmon_index"
  outdir = "results"
}

```

Save and exit this screen.

5.3 Running Nextflow

Great, now that the environment is set up, it is time to run the pipeline. Load modules Salmon and Nextflow, then run:

```
nextflow run rnaseq_salmon.nf
```

If permission is denied, try:

```

chmod +x rnaseq_salmon.nf
nextflow run rnaseq_salmon.nf

```

This will create one folder per sample in results/ with a file called quant.sf.

5.4 Merging Files

As aforementioned, there is an individual quant file for each sample. We want to merge these together in R using tximport. Begin by loading R and entering the work environment.

```

module load R
R

```

Next, enter this script.

```
# Install if needed:
# install.packages("BiocManager")
# BiocManager::install("tximport")

library(tximport)

# Get all quant.sf file paths
samples <- list.files("results", full.names=TRUE)
files <- file.path(samples, "quant.sf")
names(files) <- basename(samples)

# Import and merge
txi <- tximport(files, type = "salmon", txOut = TRUE)

# Save as TSV
write.table(txi$counts, file = "salmon_merged_counts.tsv", sep = "\t",
quote = FALSE)
```

Finally, exit R using command line:

```
q()
```

Now there is one final merged file of all the merged quant counts called "salmon_merged_counts.tsv" which is ready for analysis.

6 Appendix

6.1 Basic Shell Command Glossary

6.2 Helpful References

- <https://www.uvm.edu/vacc> — UVM VACC Help Page. The VACC are very helpful and responsive, you can send them an email and set up a meeting if needed as well.
- <https://www.educative.io/blog/bash-shell-command-cheat-sheet> - More common bash commands.
- <https://www.arch.jhu.edu/short-tutorial-how-to-create-a-slurm-script/> - How to create a Slurm Script
- <https://github.com/COMBINE-lab/salmon> — Salmon GitHub repository
- <https://www.nextflow.io/docs/latest/index.html> — Nextflow official documentation

Command	Description
<code>pwd</code>	Print the current working directory to show where you are working.
<code>ls</code>	Lists folders and files within the working directory you are in.
<code>cd foldername</code>	Change to a different directory (foldername).
<code>mkdir foldername</code>	Makes a new directory called "foldername".
<code>nano filename.sh</code>	Open a file in the Nano text editor to view or edit its contents.
<code>cat filename.txt</code>	Reads filename.txt contents and prints in the terminal.
<code>bash script.sh</code>	Run a shell script named <code>script.sh</code> .
<code>module load modulename</code>	Load a software tool on a cluster.
<code>rm file.txt</code>	Delete a file named <code>file.txt</code> .
<code>rm -r foldername</code>	Delete a folder and everything in it.

Table 1: Common in-line shell commands used in this RNA-seq preprocessing SOP.

- https://hbctraining.github.io/Intro-to-rnaseq-hpc-02/lessons/06_alignment_salmon.html — Salmon tutorial from Harvard Chan Bioinformatics
- <https://bioconductor.org/packages/release/bioc/html/tximport.html> — tximport documentation (Bioconductor)